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Mitigating Stuck Pipe by Predicting Filter Cake Thickness Using Artificial Intelligence

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Abstract

The primary objective of this paper is to mitigate stuck pipe incidents by accurately predicting filter cake thickness using advanced Artificial Intelligence (AI) techniques. A thin, impermeable filter cake is crucial for minimizing formation damage and preventing costly operational delays caused by stuck pipe issues. This study explores the application of AI models to estimate filter cake thickness using readily available drilling fluid properties. This work employs three AI-based techniques—Artificial Neural Networks (ANN), Fuzzy Logic (FL), and Functional Networks (FN)—to predict the filter cake thickness. Data for model training and testing were obtained from experiments conducted using real-field oil-based mud samples with CaCO_3 as a weighting agent. Input parameters include apparent viscosity, plastic viscosity, yield point, and drilling fluid density. The models were evaluated using different data splits for training and testing, and their accuracy was assessed using the correlation coefficient (R^2) and the arctangent absolute percentage error (AAPE). The results demonstrated strong prediction capabilities across all AI models, with R^2 values exceeding 0.84 and AAPE below 5%. Among the models, the Functional Network (FN) outperformed others, achieving an R^2 of 0.91 for training and 0.92 for testing, with corresponding AAPE values of 2.8% and 3.9%, respectively. These findings indicate that AI-driven models can provide reliable estimates of filter cake thickness, enabling proactive measures to mitigate stuck pipe incidents. This approach offers significant cost-saving potential by preventing unplanned downtime and optimizing drilling operations. This paper presents a novel application of AI techniques, specifically Functional Networks, for predicting filter cake thickness in real-field scenarios. By utilizing common drilling fluid properties, this study provides a cost-effective, data-driven approach to mitigating stuck pipe risks, offering new insights and practical solutions for the petroleum industry.

Introduction

Drilling mud is often referred to as the "blood" of drilling operations due to its essential role in maintaining wellbore stability and enabling safe drilling. It performs several critical functions, including balancing

formation pressure, preventing kicks, and cooling and lubricating downhole equipment (Hossain and Al-Majed, 2015). An additional key function of the drilling fluid is to form a filter cake, which helps minimize solid invasion into the formation and reduces formation damage. An effective filter cake should be thin, impermeable, and strong enough to withstand high differential pressure (Bageri et al., 2017). The filter cake forms as a result of the size mismatch between the solid particles in the drilling fluid and the pore openings of the formation. These particles, commonly referred to as bridging materials, accumulate at the borehole wall, creating a physical barrier (Mahto and Sharma, 2004). If the particles are too fine, they may invade the formation matrix, leading to pore blockage and a reduction in well productivity. Therefore, the drilling fluid system must be carefully engineered to ensure optimal filtration behavior, and key parameters such as filter cake thickness must be evaluated during the design and testing phases.

The properties of the filter cake are influenced by the characteristics of the mud, formation properties, and reservoir conditions. Filter cake thickness, in particular, is sensitive to factors such as rock permeability, mud rheology, and shear rate (Jiao and Sharma, 1993). Similar conclusions were drawn by Dangou et al., 2009, who demonstrated that filter cake thickness is also dependent on shear rate under dynamic filtration conditions. Various techniques are available for measuring filter cake thickness, including direct methods such as calipers, dial gauges, laser measurement systems, and computed tomography (CT) scans. Each method comes with its own advantages and limitations, as summarized by Bageri et al., 2013. However, in field applications, accurately measuring filter cake thickness is often challenging. This makes predictive modeling an attractive alternative for estimating cake thickness under operational conditions.

Artificial intelligence (AI) has emerged as a powerful tool for modeling and predicting complex behaviors in the oil and gas industry. AI techniques have been successfully applied to estimate a wide range of reservoir and fluid properties, such as permeability and porosity (Ali et al., 2013; Elkatatny et al., 2018a, 2018b), water saturation (Bageri et al., 2015), oil viscosity (Al-khdheawi, 2019; Oladiipo et al., 2009), bubble point pressure (Adeeyo, 2016), and so. hese methods offer high accuracy, speed, and the ability to handle large datasets. In drilling fluid applications, AI has also shown promise. For example, Gowida et al., 2019 developed an artificial neural network (ANN) model to predict properties of calcium chloride brine-based drilling fluids, including plastic viscosity, apparent viscosity, yield point, flow behavior index, and consistency index. Using 515 data points and input parameters such as mud weight and Marsh funnel viscosity, the model achieved a correlation coefficient greater than 0.97 and an average absolute percentage error of 6.1%. Golsefatan and Shahbazi, 2019 applied ANN, ANFIS, and least-squares support vector machine (LSSVM) models to predict filtration volume based on input variables including drilling fluid properties, KCl concentration, and SiO₂ concentration. All models delivered high accuracy, with the lowest correlation coefficient reported at 0.99. In a subsequent study, the same authors used an ANN model to evaluate the effect of various nanoparticles on filtration volume. The model achieved a correlation coefficient of 0.992 and an average absolute relative error of 2.6% (Golsefatan and Shahbazi, 2020). Oguntade et al., 2020 implemented ANN models to predict both filtration volume and filter cake thickness using inputs such as yield point, plastic viscosity, and apparent viscosity. Based on approximately 100 data points obtained from water-based mud, two ANN configurations—one with 15 neurons and another with 8 neurons in the hidden layer—were evaluated. The models achieved correlation coefficients of 0.922 and 0.999, respectively, highlighting their reliability and potential for practical application. The objective of this study is to develop AI-based predictive models for estimating filter cake thickness from measurable drilling fluid properties. This approach aims to provide an efficient and accurate alternative to direct measurement methods, particularly in field settings where such measurements are difficult or impractical.

Methodolgy

This part of the project is discussing the methodology followed in two sections, data preparation, AI techniques, and performance criteria.

Data preparation and analysis

The data was collected experimentally from oil-based drilling fluid with calcium carbonate (CaCO_3) as weighting material which used to drill sandstone horizontal section. It is around 107 data points including filter cake thickness, drilling fluid density, apparent viscosity, plastic viscosity and yield point. Mud density is one of the important properties of the mud, it can be measured using the mud balance, Usually, it calibrated using fresh water which has a density of 8.33 Ib/gal room temperature. The rheological properties of the mud can be measured using a rotational viscometer, which operate under the principle of measuring shear stress and rate. These properties are apparent viscosity (μ_a), the plastic viscosity (PV) and the yield point (YP). The relations between them as the following:

$$\mu_a = \frac{300 \theta_N}{N} \quad (1)$$

$$\mu_a = \frac{\theta_{600}}{2} \quad (2)$$

$$PV = \theta_{600} - \theta_{300} \quad (3)$$

$$YP = \theta_{300} - PV \quad (4)$$

where μ_a , θ_N , N , θ_{600} , PV , θ_{600} , and YP are apparent viscosity, dial reading, rotor speed, dial reading at 600 rpm, plastic viscosity, and dial reading at 300 rpm.

Statistical analysis of the available data is shown [Table 1](#), containing the minimum, maximum, means, median, mode, variation, kurtosis, skewness, and standard deviation for all the preparties. Moreover, [Table 2](#) shows the correlation coefficient (R^2) between the different parameters, the Coefficient analysis is used to measure the statistical correlation between the input and output values. It shown that there almost no relations between the inputs and output which is the filter cake thickness. Also, the histogram for each input and the output were performed to see the dominated range for each parameter. For the density, most of the data fell in the range of 10.25 to 11.15 as shown in [Figure 1](#). Apparent viscosity, plastic viscosity, and yield point fall in close range to each other. This is expected due to the relation between them and their histograms are shown in [Figure 2](#) to [Figure 4](#). The output that needs to be predicted is the filter cake thickness, the range of 0.177 to 0.194 represent almost 50% of the values as shown in [Figure 5](#).

Table 1—Statistical description of the data

	Density	Apparent viscosity	Plastic viscosity	Yield point	Cake thickenss
Min	10.25	52.31	44.48	13.47	0.16
Max	12.29	138.00	92.00	68.00	0.27
Arith. Mean	11.16	65.81	53.51	23.13	0.19
Geom. Mean	11.14	63.37	52.21	20.09	0.19
Mode	10.50	117.50	52.00	20.00	0.19
Standered Dev.	0.60	20.75	13.28	15.00	0.02
Median	10.94	56.35	47.44	16.24	0.19
Skewness	0.486	1.839	1.777	1.793	1.398
Kurtosis	-1.168	2.0439	1.707	1.684	2.532

Table 2—Correlation Coefficient

	Density	Apparent viscosity	Plastic viscosity	Yield point	Cake thickness
Density	1	-0.50965	-0.507761	-0.516931	0.013914
Apparent viscosity	-0.509650	1	0.994894	0.988647	0.147253
Plastic viscosity	-0.507760	0.994894	1	0.97643	0.156932
Yield point	-0.472067	0.986675	0.97643	1	0.129817
Cake thickness	0.050243	0.147253	0.156932	0.129817	1

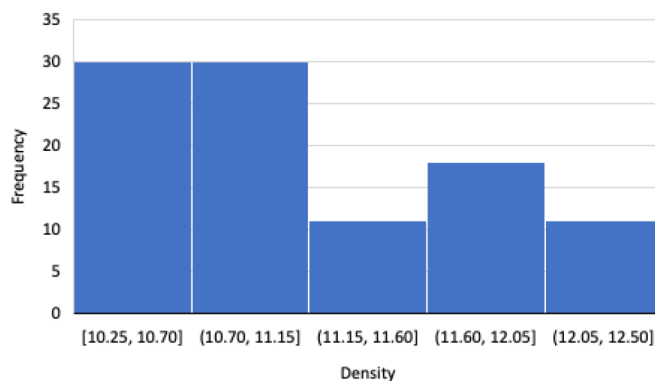


Figure 1—Histogram of the Density.

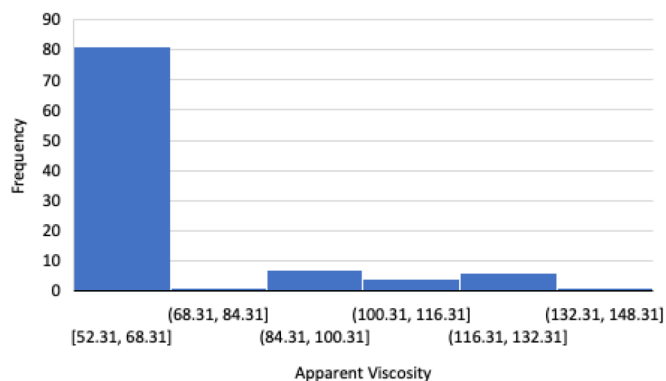


Figure 2—Histogram of the Apparent Viscosity.

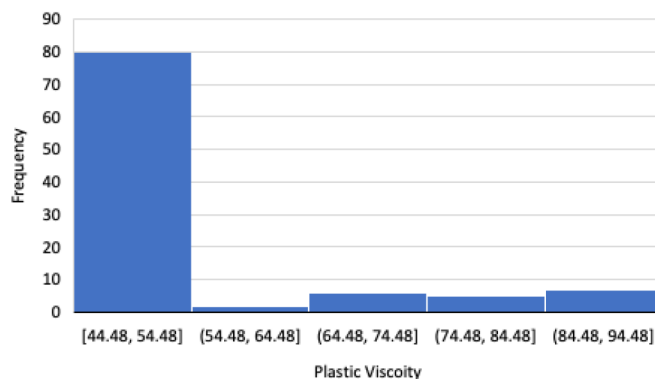


Figure 3—Histogram of the Plastic Viscosity.

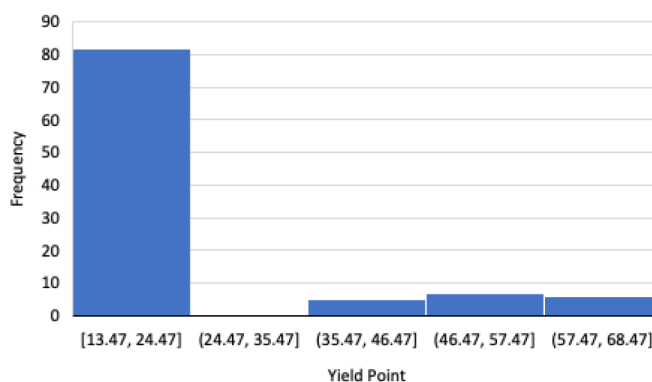


Figure 4—Histogram of the Yield Point.

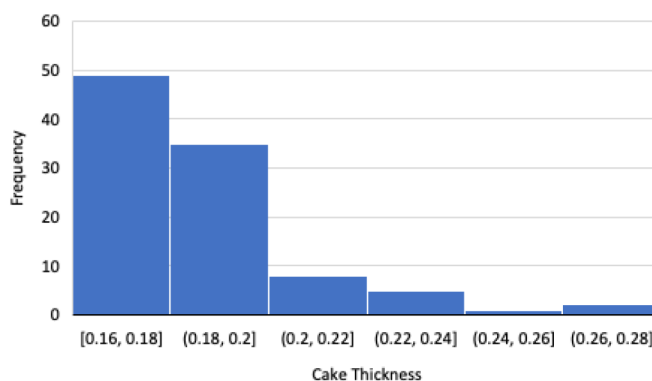


Figure 5—Histogram of the Filter Cake Thickness.

To improve the correlation coefficient, different transformation was performed for each input parameters as shown in Table 3 to Table 6. Even though there is a slight improvement in the correlation coefficient, this improvement was not carried out to the model performance. Hence, we decided to use the original data without any transformation.

Table 3—Correlation Coefficient for different transformation of density

	Density	Desnty^2	Density/2	Log(Densit y)	1/Density	Sqrt(Densi ty)	Exp(Densit y)	1/Desnity^ 2
Density	1.0000	0.99979	1.00000	0.99979	-0.999	0.99995	0.97479	-0.99808
Apparent viscosity	-0.5096	-0.49976	-0.50965	-0.51830	0.528	-0.51459	-0.40259	0.53814
Plastic viscosity	-0.5077	-0.49806	-0.50776	-0.51616	0.5258	-0.51260	-0.40201	0.53555
Yield point	-0.5169	-0.5065	-0.51693	-0.52619	0.5367	-0.52214	-0.40565	0.54721
Cake thickness	0.0139	0.0137	0.0139	0.01334	-0.0138	0.01405	0.02226	-0.01449

Table 4—Correlation Coefficient for different transformation of apparent viscosity

	Apparent viscosity	Apparent viscosity ^2	Apparent viscosity /2	Log(Apparent viscosity)	1/Apparent viscosity	Sqrt(Apparent viscosity)	Exp(Apparent viscosity)	1/Apparent viscosity ^2
Density	-0.5097	-0.4993	-0.5097	-0.5098	0.5022	-0.5115	-0.1023	0.4862
Apparent viscosity	1.0000	0.9955	1.0000	0.9960	-0.9850	0.9990	0.3514	-0.9683
PV	0.9949	0.9865	0.9949	0.9943	-0.9862	0.9956	0.2927	-0.9720
YP	0.9886	0.9801	0.9886	0.9877	-0.9787	0.9893	0.3022	-0.9631
Cake thickenss	0.1473	0.1187	0.1473	0.1879	-0.2371	0.1660	0.1848	-0.2915

Table 5—Correlation Coefficient for different transformation of plastic viscosity

	PV	PV ²	PV /2	Log(PV)	1/PV	Sqrt(PV)	Exp(PV)	1/PV ²
Density	-0.5078	-0.5020	-0.5078	-0.5065	0.5005	-0.5086	-0.1305	0.4883
Apparent viscosity	0.9949	0.9932	0.9949	0.9913	-0.9827	0.9938	0.4226	-0.9693
PV	1.0000	0.9974	1.0000	0.9974	-0.9898	0.9994	0.3723	-0.9775
YP	0.9764	0.9711	0.9764	0.9760	-0.9699	0.9770	0.3642	-0.9586
Cake thickenss	0.1569	0.1244	0.1569	0.1966	-0.2408	0.1757	0.1828	-0.2879

Table 6—Correlation Coefficient for different transformation of yield point

	YP	YP ²	YP /2	Log(YP)	1 YP	Sqrt(YP)	Exp(YP)	1/ YP ²
Density	-0.5169	-0.5056	-0.5169	-0.5035	0.4600	-0.5144	-0.1132	0.3891
Apparent viscosity	0.9886	0.9787	0.9886	0.9797	-0.9470	0.9869	0.3760	-0.8928
Plastic viscosity	0.9764	0.9590	0.9764	0.9748	-0.9485	0.9785	0.3162	-0.8991
Yield point	1.0000	0.9912	1.0000	0.9893	-0.9547	0.9975	0.3299	-0.8988
Cake thickenss	0.1298	0.0789	0.1298	0.2182	-0.3296	0.1694	0.1809	-0.4372

AI techniques

Different AI methods were used in this work including artificial neuron network (ANN), Fuzzy logic (FL), functional network (FN), and Supported vector machine (SVM). Here a brief description and implementation for each method.

ANN. ANN is method that tries to imitate how the biological nervous system work, to find a relation between input and output. The implantation of the ANN in this work has been done using MATLAB software, different trails were carried out to achieve this best results. This was performed by varying seed number, ratio of the data used for training and testing, number of neurons, training functions, testing functions, and network function. Seed number which represents the randomness was varied arbitrary, two data ratios were used including 70:30 and 80:20. The training functions investigated including trainlm, trainr, triainoss, trainbr, traingdx, trainscg, traincgb, traingda, and trainbfg. The investigated testing function include softmax, logsig, poslin, hardlim, radbas, satlins, netinv, and tribas. For the network functions, two were implemented including newlrn and newfit.

FL. FL method is a reasoning method instead of precession. An example for it is fuzzy group theory that have different objectives and boundaries where membership is a problem of degree. Two different tool of fuzzy logic were implemented in this work namely Mamdani FIS (Genfis 1) and Sugeno FIS (Genfis 2). Like the ANN, different parameters were varied to reached to the best outcomes including seed, epoch size,

radii, data ratio, input and output membership functions. The investigated membership used are gaussmf, trimf, and newbr while the output membership used ere linear and constant.

FN. FN considered as an extension of the ANN with different that it has several layers of neurons and each processing neuron makes an explicit calculation. The best results were obtained by trial-and-error method, different parameters were varied including the methods, functional type and data ratio. The studied methods are functional network forward-backward method (FNFBM), functional network backward-forward method (FNBFM), functional network backward-exhaustive method (FNBEM), functional network exhaustive selection method (FNESM), and functional network forward-selection method (FNFSM). The investigated types are linear, non-linear(1), non-linear(2), and non-linear(3).

Results & Discussion

The different AI methods gave varied results, since some method suit the problem better compared to the other. The accuracy of the model was assessed using correlation coefficient for the training (CC) and testing plus the average absolute percent error (AAPE). For the ANN, different runs were performed by varying different parameters. The fifth run seems the best overall with correlation coefficient of 0.93 for the training and testing phases. The AAPE for the training phase equal to 2.9% while for the testing phase equal to 3%. This was achieved by using 6 neurons, newlrn as network function, trainlm as training function, softmax as testing function and the ratio was 70% for the training and 30% for the testing. The results of this run are shown in Figure 6.

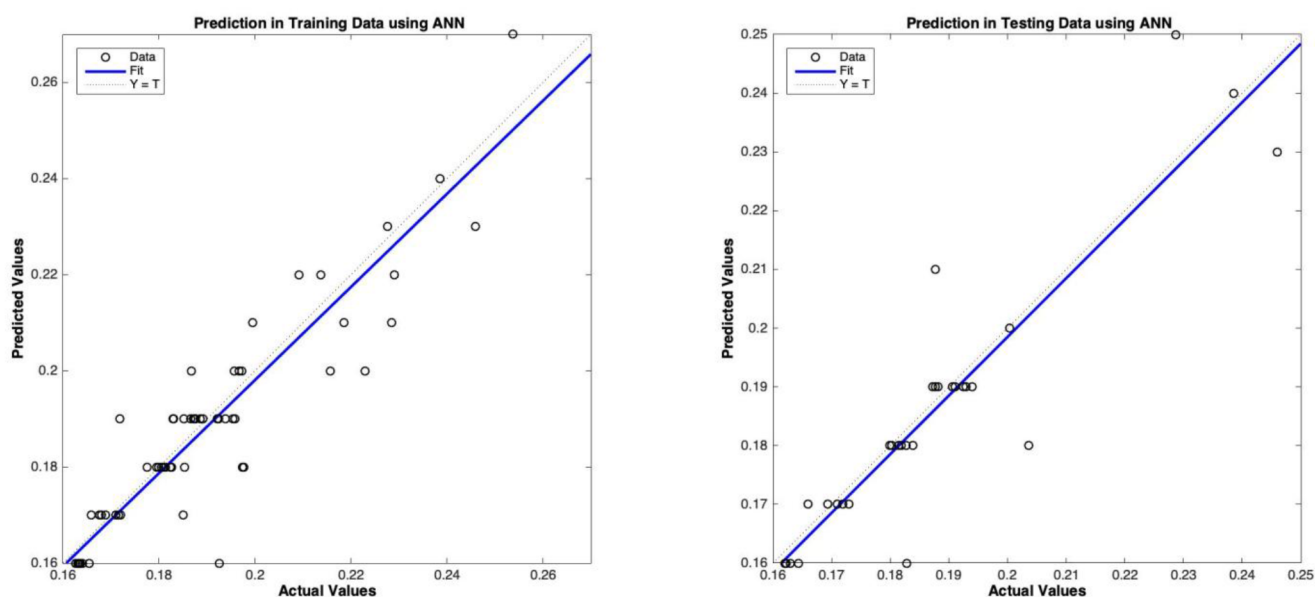


Figure 6—Results of the best ANN run in the training and testing stages.

In addition, the implementation of the FL showed good results as well. The best results are found at epoch size of 200, radius of 0.7, Genfis 1 as FL tool, newrb and constant as input and output memberships, respectively. The FL model predicted the filter cake thickness in the training set (80% of the data) with CC equal to 0.9632 and AAPE of 2.2832. Also, the FL model was able to predict the filter cake thickness in the testing stage (20% of the data) with CC of 0.8431 and AAPE of 4.9243. The results can be show in Figure 7.

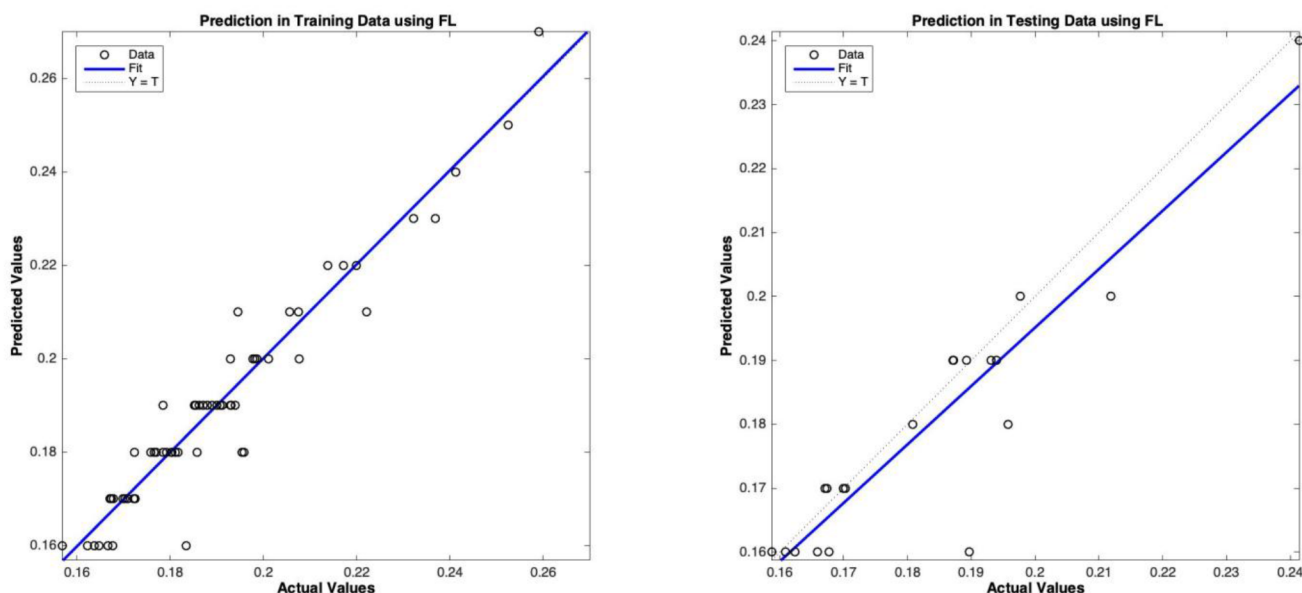


Figure 7—Results of the best FL run in the training and testing stages.

The results from the FN are shown in Figure 8, the best results for this method were obtained by using FNESM method, Non-linear (2) as type, and 75% of the data used for the training while the remaining for testing. The training stage produced good accuracy with CC of 0.9081 and AAPE equal to 2.834. In the testing step, the prediction was excellent with CC of 0.9194 and AAPE equal to 3.9594.

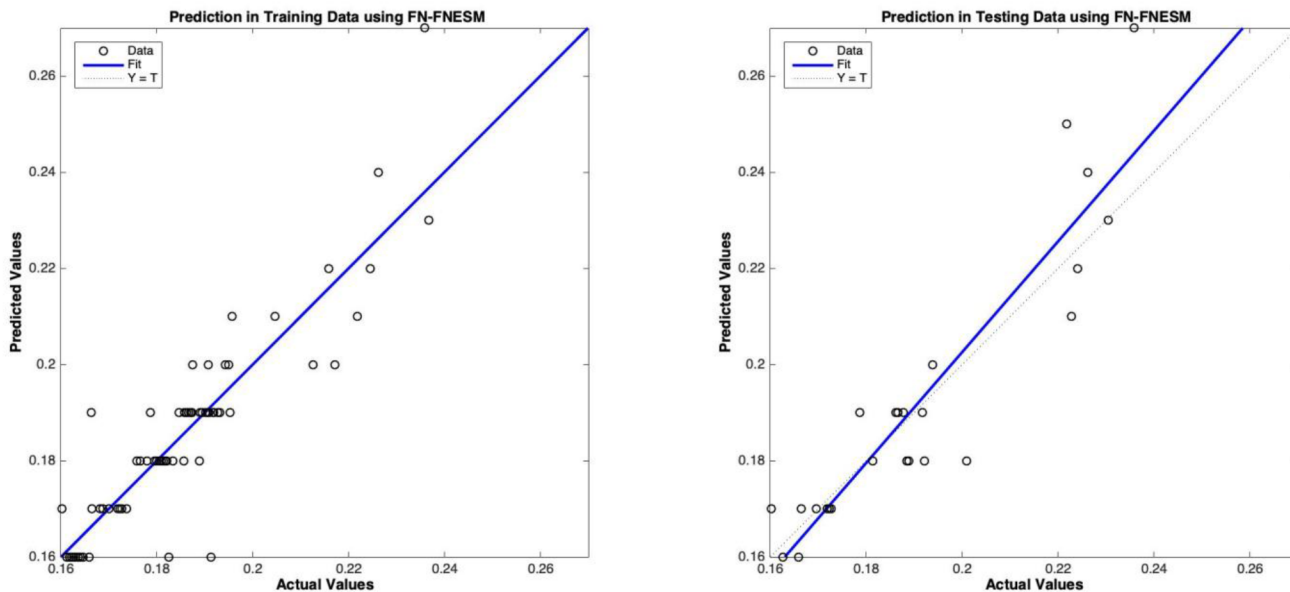


Figure 8—Results of the best FN run in the training and testing stages.

Comparison between AI Techniques

A comparison is made between the used AI techniques to select the best among them based on the prediction accuracy. The best results criteria of each technique (i.e., CC training, CC testing, AAPE training, and AAPE testing) are shown in Figure 9. The performance of the different methods is relatively similar, but it seems that the FN has small edge overall compared to the other techniques.

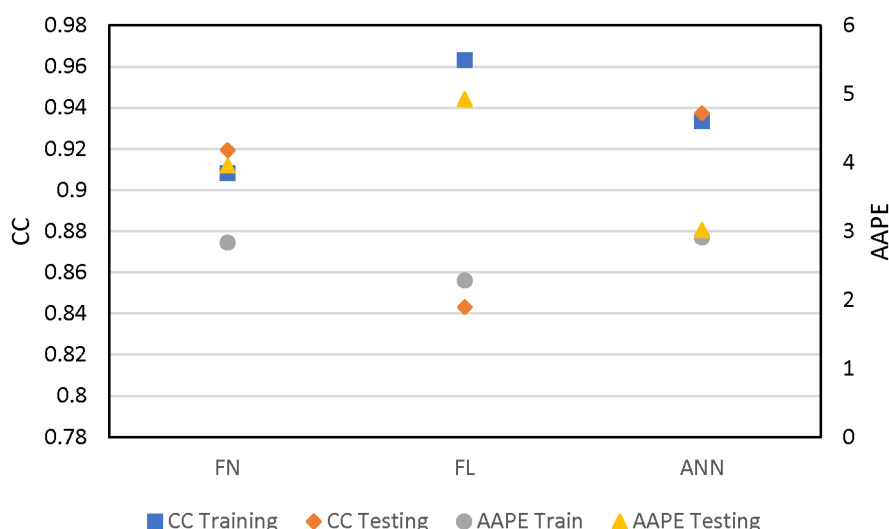


Figure 9—Comparison of the three AI techniques for the training and testing datasets.

Conclusion

The filter cake thickness is essential properties that need to be determined accurately the AI techniques were utilized to achieve this objective. The AI techniques that were implemented in this study include ANN, FL, and FN. All AI techniques showed good predication of the filter cake thickness with the best performance obtained through the usage of the FN with CC equal to 0.9081 and 0.9194 for training and testing phases, respectively. Also, the training AAPE equal to 2.83% while the testing AAPE equal to 3.95%.

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